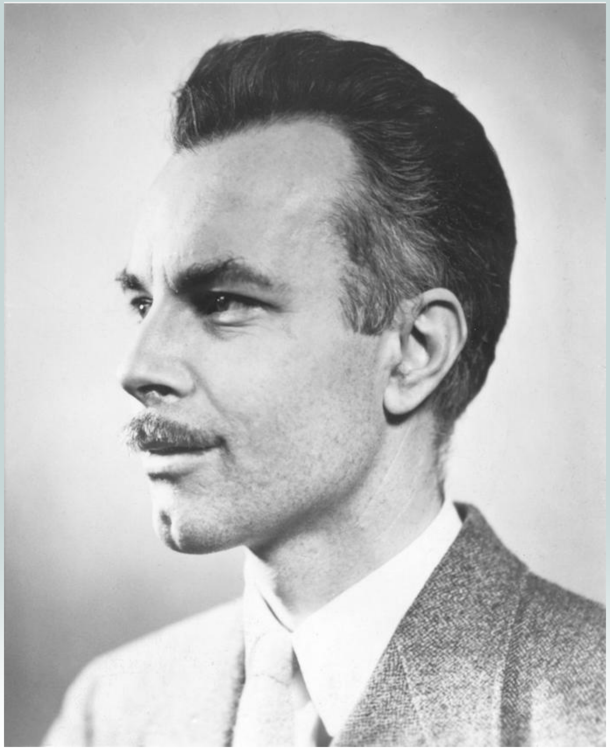


George Arthur Philbrick

Philbrick operational amplifier, circa 1952

Lecture 4: Operational Amplifiers (Op Amps)

Saturation-2

What's happening is that the op amp is saturating, that is, moving out of the linear region and ceasing to operate as an amplifier. We can only model the op amp as ideal when it is not saturated.

We can check saturation, though, by comparing output voltage and current to saturation limits:

$$|v_o| \le v_{sat}$$

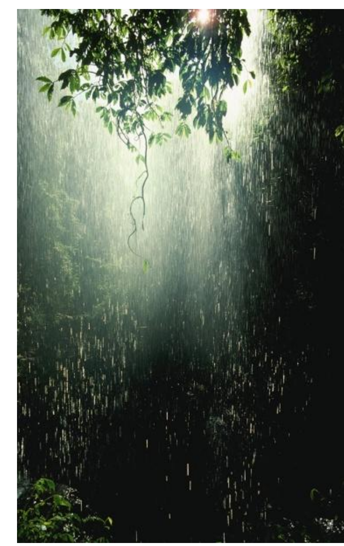
$$|i_o| \le i_{sat}$$

v_{sat} and i_{sat} come from the specification sheet. v_{sat} is usually a volt or two less than the power supply voltage.

Saturation-3

For example, if the power supply is +/- 12V and i_{sat} is 2mA, then the maximum output voltage is 10 V if Vsat is 10V.

If Vsat is 10V, what's the minimum RL?



Saturation-3

For example, if the power supply is +/- 12V and i_{sat} is 2mA, then the maximum output voltage is 10 V if Vsat is 10V.

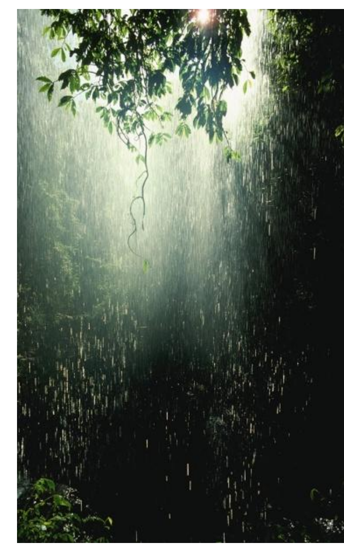
If Vsat is 10V, what's the minimum RL?

$$i_o \le i_{sat}$$

$$\frac{v_o}{R_L} \le 2mA$$

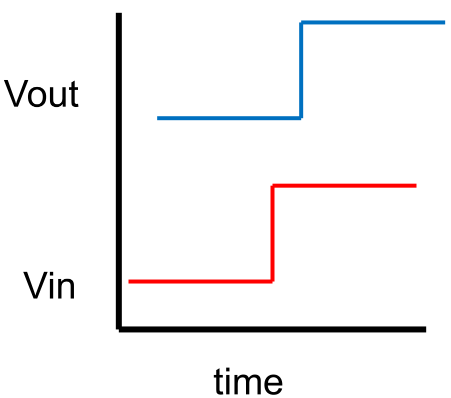
$$\frac{10}{R_L} \le 2mA$$

$$R_L \ge \frac{10}{2\times 10^{-3}} = 5k\Omega$$

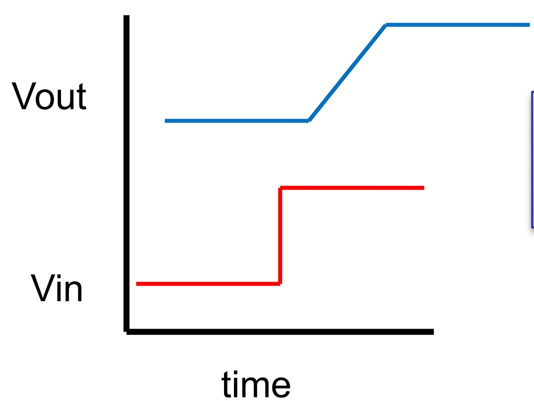


Slew rate

At this point, we should probably note that another departure from ideality in the op amp is that the slew rate, $\frac{dv_o}{dt}$, is also limited. This affects how fast the op amp responds to a step change in input, and how high a frequency it can operate at. We won't pay much attention to slew rate in this course, but you will run into it in later courses, and in the real world!



Ideal op amp

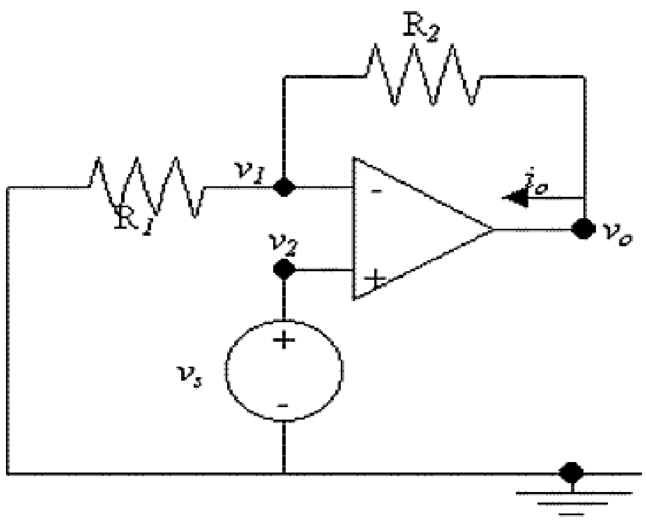


Real op amp

The slew rate for the 741 op amp is 0.5V per microsecond

More analysis

Let's look at another circuit (you should try the analysis)



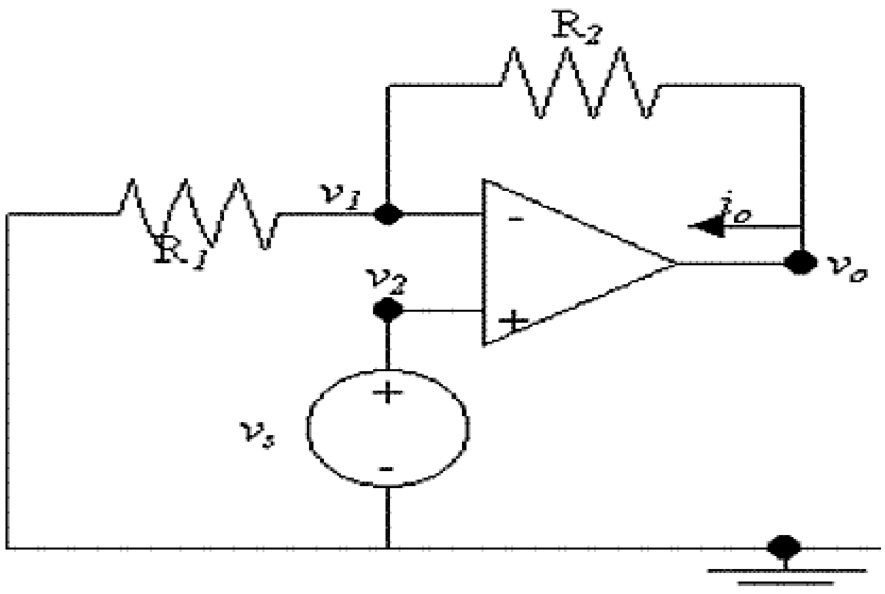
As before, find v_o as a function of v_{os}.

First, what's v_2? v_s.

So, what's v_1? v_s

This is a good time to practice active learning. Stop here and try to solve for v_0 as a function of V_{os} . You can always advance to the next slide and see the answer, but that's no fun!

More analysis-2



Now, write KCL at v_1:

$$\frac{v_s}{R_1} + \frac{v_s - v_o}{R_2} = 0$$

$$v_s\left(\frac{1}{R_1} + \frac{1}{R_2}\right) - \frac{v_o}{R_2} = 0$$

$$v_o = v_s\left(1 + \frac{R_2}{R_1}\right)$$

Key term

- Potentiometer

A three-terminal resistor with a center sliding contact (wiper) that forms a voltage divider or a variable resistor

This is called a non-inverting amplifier. Notice that the gain of the amplifier can be changed by changing the ratio of the resistors, for example with a potentiometer for R_2.

Design with op amps

Somewhere, in your textbook* (or some textbook somewhere!), is a catalog of op amp circuits that perform different functions. There are lots, beyond what's in the book. The circuits are generic; that is, they do not have specific values for the resistances. The catalog is a starting point for circuit design. We now know enough to do some almost real-world design. The process is:

- Given a function to perform
- Choose a circuit from the catalog that performs the function. (Or, invent a new one. If you do, patent it.)
- Figure out values for the circuit components so that the specific function we want is implemented (e.g., specific gain).
- Check the circuit against limits.
- Iterate as required.

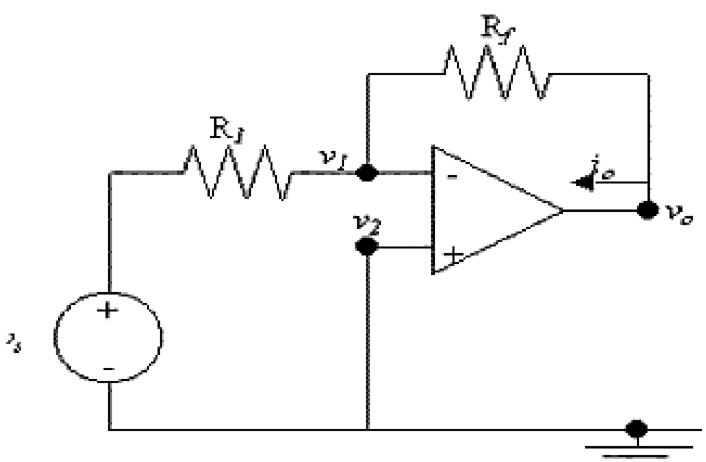
* www.stanford.edu/class/ee122/.../**op_amp_circuit**%20collection_AN-31.pdf

Design with op amps-2

Example: We want to amplify an input signal by a factor of 100 (gain of 100). The input signal has a range of 0 to -100 mV, but we want a positive voltage output.

Pretty clearly we want the inverting amplifier circuit from the catalog:

Please wait while I thumb through the catalog. OK, I found it:



Let's check the analysis:

$$v_2 = 0$$

$$v_1 = 0$$

$$\frac{v_s - 0}{R_1} + \frac{v_o - 0}{R_f} = 0$$

$$v_o = -\frac{R_f}{R_1} v_s$$

Robert Widlar inventor of the first mass-produced op amp, uA709

Design with op amps-3

For our design we want $\dfrac{R_f}{R_1} = 100$

but, that leaves a lot of choice. What other concerns are there?

Source current: Suppose the source is limited to 10 mA output current.
(This is a new part of the specification!) Then

$$\frac{v_s}{R_a} < 10mA$$

$$\frac{100mV}{10mA} < R_1$$

$$R_1 > 10\Omega$$

$$R_f = 1K\Omega$$

Design with op amps-4

Is this a good design?

What about v_{sat}? $v_{o\max} = -100 \cdot (-100mV) = 10V$

We'll need a power supply of around +/- 11V. +/- 12 V is a common standard (you can find this voltage inside many desktop PCs) or +/- 15V, both are OK but NOT +/- 5V, which is the most common digital electronic power supply voltage.

What about i_{sat}? $i_o = \frac{v_o}{R_f} = \frac{10}{1K} = 10mA(!)$

Whoops, the limit we talked about earlier was 2 mA. (This limit will vary from op amp type to op amp type, so maybe we just need to buy a different op amp. But let's use the one in the parts kit... with a 2mA output current limit - another new part of the specification.)

Design with op amps-5

If we increase *Rf*, and increase *R1* to keep the 100 ratio, then i_S will also go down, and we will meet both constraints.

$$i_o < 2mA$$

$$\frac{v_o}{R_f} < 2mA$$

$$R_f > \frac{10}{2mA} = 5K\Omega$$

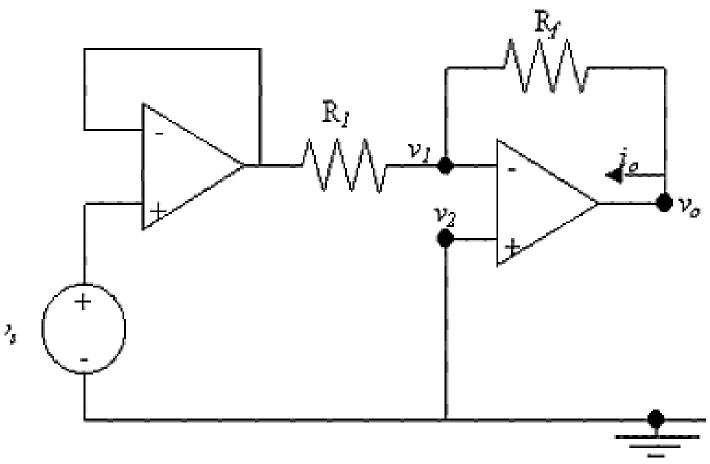
$$R_1 = \frac{R_f}{100} = 50\Omega$$

Note that this also meets the general guideline to use 5KΩ – 10KΩ resistors around the op amp.

Design with op amps-6

We should probably also worry about resistor power dissipation, but we will leave that up to you.

What could we do if the source current limit was much, much smaller?



We could cascade op amps, using a voltage follower then an inverting amplifier, for example.

Real op amps

The ideal op amp has already been brought closer to reality by adding saturation and slew rate limits, but the core component is still an ideal VCVS.

In real op amps the VCVS is not ideal. In particular, real op amps have:

- a finite (but large) gain, 200,000 versus infinite in the ideal model
- a finite (but large) input resistance, 2MΩ versus infinite
- a small output resistance, 75Ω versus zero
- tiny bias currents, 80 nA versus zero
- small offset voltage, 1 mV versus zero

Checkpoint

That’s pretty much it for op amps until we know something about energy storage devices.

Op amp problems are a new type of circuit problem. Like other types of circuit theory problems, practice is the best way to improve your solution abilities.

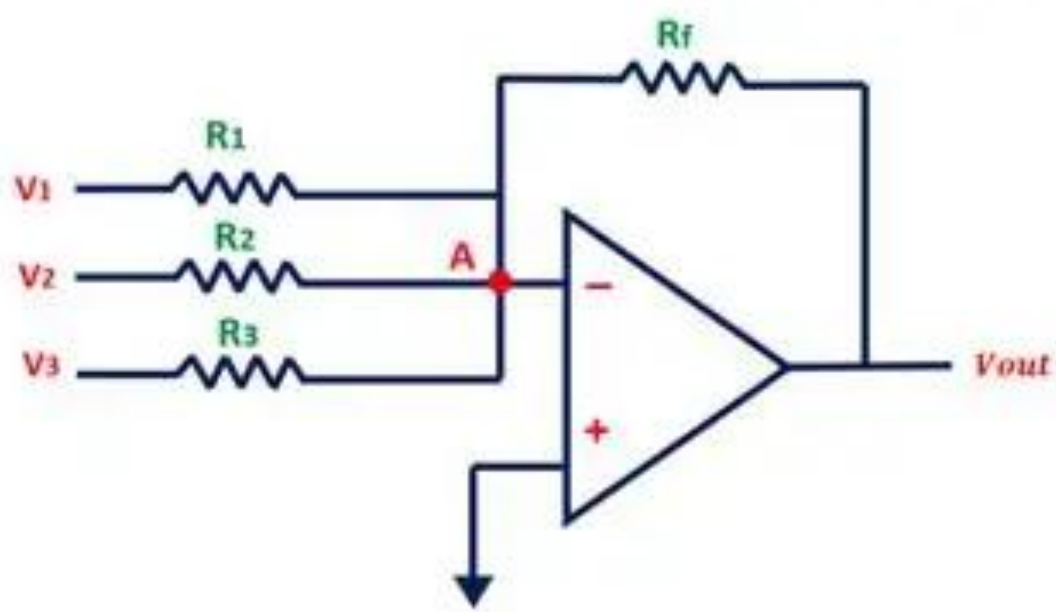
In particular, try to design some real circuits. Create a set of specifications, choose an op amp, and then use published data from the a manufacturer’s data sheet and do some calculations for the non-idea case to see how close your design meets specs.

Of course, there is a lot about op amps that we’ve yet to consider. We haven’t look at issues such as temperature dependence of the specs, sensitivity to power supply voltage issues and circuit stability, to name a just a few!

These will come. If not in this class, certainly in the classes to follow. Stay tuned!

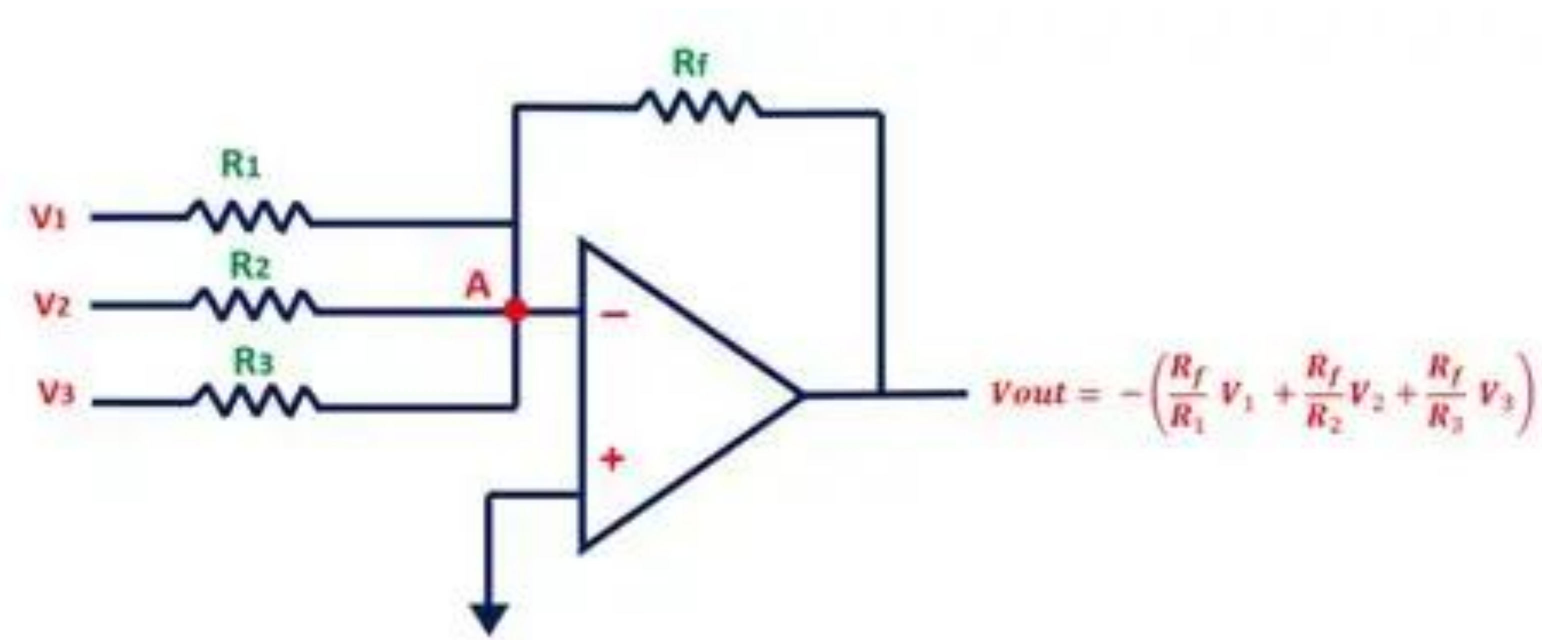
Summer

A superposition of multiple inverter - What is Vout for the following circuit:



Summer

A superposition of multiple inverter:



Summary

In this lesson you learned:

- What the op amp symbol means
- The special cases that allow us to do op amp circuit analysis
- Several types of op amp circuits
    - Voltage follower
    - Inverting amplifier
    - Non-inverting amplifier
- How to set the circuit parameters and check the design for saturation
- How to calculate circuit parameters using more realistic models for the op amp models